DECAMP
Open Distributed European Virtual CAMPus on ICT Security
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Metropolia

Certificate

Badea Theodor Julian
Bachelor student
Politehnica of Bucharest
has completed the DECAMP course :

Applied Web Application Security

during the semester 2024-2025 and obtained a performance evaluation of **87.67/100*.** (*The grade will be calculated at the home university based on its local scale.)

The student passed an online presence examination in her/his home university given and corrected by the course instructor. According to the agreement of all partner universities within the DECAMP Project Consortium the student is entitled to obtain 6 ECTS from her/his home university. The course corresponds to a **learning effort of** a minimum of 150 hours/semester.

A short description of the course content is listed on the back of this certificate.

The course was taught by Kimmo Saurén from the Metropolia University of Applied Sciences, Finland.





17.6.2026
Date Signature

DECAMP *stands for Open Distributed European Virtual CAMPus and has been set up for the purpose of delivering specialist courses focused on ICT Security.* **DECAMP** *is a unique and unprecedented international strategic partnership between six EU IT faculties of well-known EU universities in Finland, Germany, Italy, Romania, Spain and the UK.* **DECAMP** *is co-funded by the ERASMUS+ Programme of the European Union. Grant no. 2014-1-DE01-KA203-000695.*

Learning objectives:

After completing this course students will be able to:
- Be aware of the most severe WEP application attacks occurring daily world-wide (OWASP TOP-10).
- Evaluate the impact of organizational risks associated with different WEB application threads
- Use the Penetration Testing Execution Standard as a guideline to create organized and well documented WEB application testing procedure.
- Apply techniques and tools to test practical WEB application security level.
- Repair practical WEB application security vulnerabilities revealed by penetration testing.
- Apply in practice the principles recommended for designing, developing and implementing secure Web applications.

Short content of the course (Bachelor level):

***Chapter 1*: Introduction to WEB Application Security :** Importance of the web applications, Required preliminary knowledge, HML, SQL, Javascript, Legal warning.
Chapter 2: **WEB Application Technologies and Frameworks:** Purpose of a web-application, Frameworks, Common and Custom modules for frameworks, Tailored web-applications for specific purposes.
***Chapter 3*: WEB Application Defense Mechanisms:** Handling user access, Handling user input, Controlling web application changes, Managing the application, Handling attacks.
***Chapter 4*: Mapping the Application:** Using common resources for reconnaissance, Detecting implementation technology, Detecting content and functionality, Automated web application mapping.
***Chapter 5*: By-passing Client Side Controls:** Inspecting client side data, Manipulating form fields, Detecting hidden data.
***Chapter 6*: Attacking Authentication:** Associated functionalities of Authentication, Securing Authentication, Authentication weaknesses, Preventing brute forcing, Preventing information leakage.
***Chapter 7*: Attacking Session Management:** Session management in web applications, Weaknesses in session token generation, Weaknesses in the handling of session tokens,
***Chapter 8*: Attacking Access Controls:** Web application access control mechanisms, Detecting unprotected functionality, Direct access methods to API, Identifier based functions, Multistage functionality.
***Chapter 9*: Injecting code:** SQL-injections, Basic SQL-injection, Response-based SQL-injections (union, blind, error based), Data extraction-based SQL-injections, Impact point-based SQL-injections, Locating and fixing SQL-injections.
***Chapter 10*: Attacking Other Users:** Detecting, using and preventing cross site scripting attacks(XSS) (reflected, stored and DOM XSSs), Cross-site Request Forgery attacks (CSRF), Combination attacks, Protecting against XSS and CSRF attacks.
***Chapter 11*: Other Exploitation Methods:** Path travelsar attack, Insecure direct object references, File inclusion vulnerabilities, Unvalidated redirects and forwards, OS command injections, Attacking application logic.
***Chapter 12*: Web Application Security Tools:** Intercepting proxies, Browser add-ons, Stand-alone applications.
***Chapter 13*: Penetration Testing Standards:** The Open Source Security Testing Methodology (OSSTM), Penetration Testing Execution Standard (PTEST), OWASP V4 Testing Guide.